

FLOATING POINT ARITHMETIC AND STABILITY

1.1 SOME DEFINITIONS

LET a BE SOME EXACT (REFERENCE) VALUE,
AND LET $\hat{a}$ BE SOME APPROXIMATION OF a (FLOATING
PT. APPROX)

ABSOLUTE ERROR (SOMETIMES SIMPLY ERROR)

$$e_{abs} \equiv a - \hat{a}$$

(OR $\hat{a} - a$, DOESN'T MATTER)

RELATIVE ERROR (FRACTIONAL ERROR)

$$e_{rel} \equiv \frac{a - \hat{a}}{|a|}$$

FO

Add to New Copilot Notebook

Rename

X

1.2 IEEE 754 64-BIT FLOATING PT ARITHMETIC

- 53 BITS MANTISSA, 11 BITS EXPONENT, BASE 2
REPRESENTATION FOR BOTH; CAN REPRESENT
NUMBERS IN THE RANGE

$$\sim 2 \times 10^{-308} \dots 1 \times 10^{308}$$

WITH TOUGHLY 16 DECIMAL DIGITS PRECISION

- $1/f_{min} < f_{max}$ SO THAT $1/f_{min}$ DOESN'T OVERFLOW
- SPECIAL (EXCEPTIONAL) VALUES

infinity EXAMPLE: $1.0 / 0.0$

nan (NOT A NUMBER) EXAMPLE: $0.0 / 0.0$

- nan IS "CONTAGIOUS" nan $[* / + -]$ ANYTHING

= now e.g.

1.3 MACHINE PRECISION / MACHINE EPSILON, ϵ

- SMALLEST $\epsilon > 0$ SUCH THAT

$$|1 \pm \epsilon| \neq 1$$

IN THE FLOATING PT MODEL

$$\text{IEEE 754} \rightarrow \epsilon \sim 10^{-16}$$

1.4 ROUND-OFF ERROR (FINITE PRECISION ERROR); CATASTROPHIC LOSS OF PRECISION (C.L.O.P.)

- MULTIPLIES / DIVIDES: NOT MUCH OF A PROBLEM.
FOR "RANDOM" OPERATIONS, EXPECT RELATIVE ERROR TO GROW LIKE $\sqrt{n}$ WHERE n IS THE NUMBER OF OPERATIONS
- ADDITION / SUBTRACTION: SERIOUS PROBLEM WHEN
SUBTRACTING x, y WITH $x \approx y$

$$\begin{array}{r} 3.6326 \\ - 3.6325 \\ \hline \end{array} \quad \begin{array}{l} 5 \text{ SIG. FIGS.} \\ 5 \text{ SIG. FIGS.} \end{array}$$

- $\ln 4$

2. STABILITY / NUMERICAL STABILITY

- CONSIDER SOME PROCESS / ALGORITHM / MAPPING

$$U(x) \rightarrow y$$

$$\hat{U}(\hat{x}) \rightarrow \hat{y}$$

$\hat{x}, \hat{y}$ TYPICALLY VECTORS



FLOATING PT,

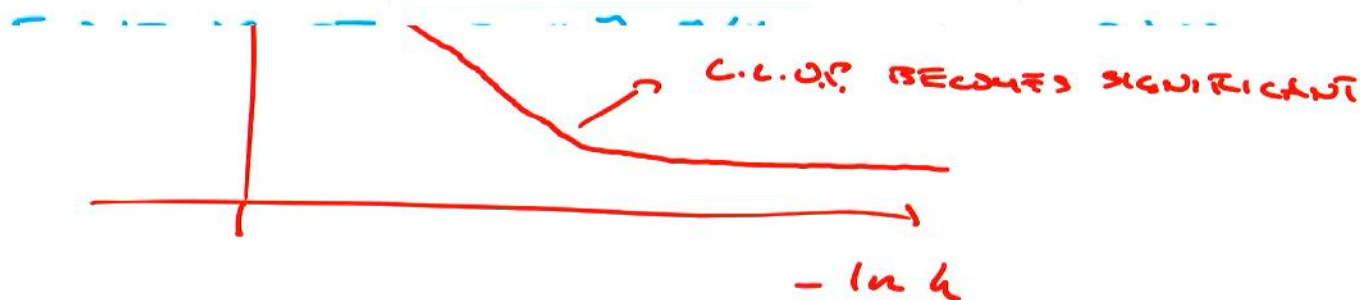
HEURISTIC (NON-RIGOROUS) DEFINITION OF STABILITY

CONTINUOUS: IF $\|x - x'\|$ IS SMALL THEN SO IS

$\|U(x) - U(x')\|$
 SOME DISTANCE
 (VALUES COULD BE DIFFERENT)

WORM, ..E.

$U(x), U(x')$



2. STABILITY / NUMERICAL STABILITY

• CONSIDER SOME PROCESS / ALGORITHM / MAPPING

$$U(x) \rightarrow \gamma \quad \text{EXACT}$$

$$\hat{U}(\hat{x}) \rightarrow \hat{\gamma} \quad \text{FLOATING PT.}$$

$\hat{x}, \hat{\gamma}$ TYPICALLY VECTORS

HEURISTIC (NO) OF STABILITY

CONTINUUM: THEN SO IS

$\|U(x) - U(x')\|$ WORM, ..E.

SOME DISTANCE BETWEEN x, x' $U(x), U(x')$
 (VALUES COULD BE DIFFERENT)

FLOATING PT: IF $\|\hat{x} - \hat{x}'\|$ IS SMALL THEN SO IS
 $\|\hat{U}(\hat{x}) - \hat{U}(\hat{x}')\|$

• UNFORTUNATE FACT; STABILITY OF $U(x)$ DOES NOT
 GUARANTEE STABILITY $\hat{U}(\hat{x})$

IF $\hat{U}$ IS NOT STABLE, ROUND OFF ERROR (FP)

WILL LIKELY RENDER THE METHOD USELESS

$\hat{U}$ IS THEN CALLED "NUMERICALLY UNSTABLE"
 "ILL-CONDITIONED"